

Fundamentals of Photovoltaics and solar cell technology

Complete student lesson for course code **6501C**.

Revision 2026 Semester 6 Open Elective 4 modules

Course snapshot

Scheme: 4 (L:4, T:0, P:0)

Credits: 4

Contact: 60 periods per semester

Module hours listed: 58

Official Source Declaration

This lesson uses the supplied official syllabus PDF as the authoritative source for course information, revision, modules, topics, objectives, Course Outcomes, assessment information, official activities, practical requirements, and references.

Source handling: Official wording and numerical values are retained wherever supplied. Educational explanations, Malayalam support, examples, diagrams, and revision questions are added as study support and are not presented as additional official requirements.

Transparency note: The supplied PDF does not specify a separate CIE/ESE distribution. Any limitation in the supplied PDF is not silently corrected.

How to Study This Lesson

Recommended sequence

1. Begin with the official source declaration and course dashboard.
2. Study each module in the official order and keep the coverage table as a checklist.
3. Open every topic, understand the example or procedure, and write your own short answer.
4. Use the revision section, glossary, questions, and practice paper after completing the modules.

Study principle: Keep English technical terminology, symbols, units, and official assessment wording unchanged in examination answers. Use the Malayalam support blocks only as a bridge to understanding.

Course Dashboard

Course code

6501C

Semester

6

Credits

4

Teaching scheme

4 (L:4, T:0, P:0)

Module-hour and outcome mapping

No.	Module	Official hours	Relevant CO / level
1	Renewable Energy Fundamentals	12	CO1
2	Solar-Cell Fundamentals	12	CO2
3	Solar-Cell Characteristics and Efficiency	16	CO3
4	Solar-Cell and PV-Module Fabrication	18	CO4

Prerequisites

- Semiconductor theory — Basic Electronics, code 2041, Semester 2.

How to study this lesson

1. Read the module introduction and learning goals.
2. Open every topic and reproduce its example, sequence, or procedure.
3. Use Malayalam support as a bridge; retain English technical terms for examinations.
4. Attempt the module questions without looking at the answers.

Objectives, Course Outcomes and CO-PO Information

Official course objectives

1. The course aims to provide a basic understanding of photovoltaics and solar cell technology.

Course Outcomes

1. Explain different renewable and non-renewable energy sources and their types.
2. Describe the fundamentals of solar cells.
3. Explain solar-cell performance and the factors affecting it.
4. Explain solar-cell and PV-module fabrication and formation.

CO-PO mapping as supplied

The supplied CO-PO table is reproduced at outcome level from the PDF; blank cells are not treated as mapped.

Legend: 3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped.

Complete Syllabus Coverage

Use this checklist to confirm that every official module topic is represented in the lesson.

Complete official topic coverage checklist

Module	Topic code	Topic	Hours
module-1	M1.01	Need of switching to Renewable Energy sources	2
module-1	M1.02	Renewable and Non-renewable sources	2
module-1	M1.03	Main sources of Renewable	4
module-1	M1.04	Applications, Advantages & Disadvantages of Renewable Energy	4
module-2	M2.01	Current conduction in semiconductor	3
module-2	M2.02	Formation of P-N junction of Semiconductor	3
module-2	M2.03	Materials for Opto-Electronic applications	3
module-2	M2.04	Concept of solar cell and its main elements	3
module-3	M3.01	Characteristics of Solar cells	4
module-3	M3.02	Design considerations of Solar cells	4
module-3	M3.03	Factors limiting the efficiency of solar cell	4
module-3	M3.04	Impact of external parameters on solar cell performances	4
module-4	M4.01	Classification of solar cells	4
module-4	M4.02	Materials used for Photovoltaic Cells	4
module-4	M4.03	Technologies for Photovoltaic Cells Fabrication	5
module-4	M4.04	Fabrication and formation of PV modules	5

Learning Roadmap

1. Renewable Energy Fundamentals
CO1 · 12 hours

2. Solar-Cell Fundamentals
CO2 · 12 hours

3. Solar-Cell Characteristics and Efficiency
CO3 · 16 hours

4. Solar-Cell and PV-Module Fabrication
CO4 · 18 hours

The roadmap follows the order of the supplied syllabus.

OFFICIAL MODULE 1

Renewable Energy Fundamentals

Photovoltaic technology is part of a wider energy system in which resource availability, environmental impact, and conversion method determine sustainability.

Hours: 12

CO1 · Bloom level: Understanding

Learning goals

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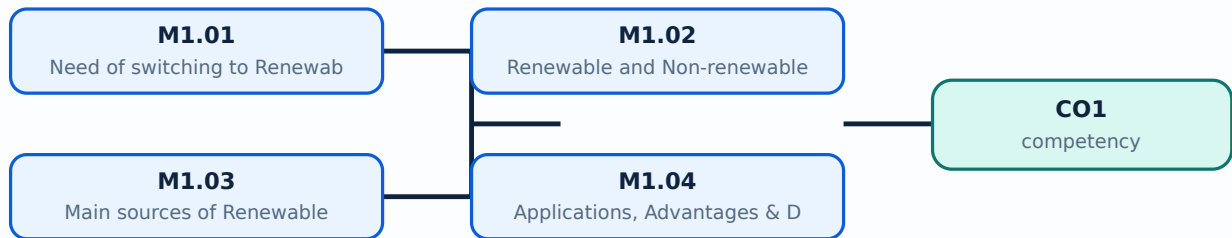
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Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-1: the listed topics build the module competency.

Concept and explanation

Scope. The official syllabus places **Need of switching to Renewable Energy sources** in this topic. The required scope is: Need for switching to renewable energy sources.

Concept and method. Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

Aspect	What to learn
Purpose	Why the device, method, material, network, or process is required in photovoltaic engineering.
Working idea	How the input is converted, controlled, transported, stored, measured, or protected.
Performance	Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.
Engineering decision	How an engineer selects a configuration, operating method, component, or maintenance action.

Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.

Malayalam support: **Malayalam support:** Need of switching to Renewable Energy sources topic purpose, working principle, components variables, performance safety checks. Technical terms English-; diagram step sequence process.

Study points

- Define Need of switching to Renewable Energy sources using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.
- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Need of switching to Renewable Energy sources is applied in photovoltaic engineering practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For ****Need of switching to Renewable Energy sources****, prepare a four-column revision note with ***purpose***, ***principle/sequence***, ***important parameters***, and ***application or limitation***. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Need of switching to Renewable Energy sources without explaining the working sequence, variables, or engineering reason for each item.

Concept and explanation

Scope. The official syllabus places **Renewable and Non-renewable sources** in this topic. The required scope is: Difference between renewable and non-renewable sources.

Concept and method. Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

Aspect	What to learn
Purpose	Why the device, method, material, network, or process is required in photovoltaic engineering.
Working idea	How the input is converted, controlled, transported, stored, measured, or protected.
Performance	Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.
Engineering decision	How an engineer selects a configuration, operating method, component, or maintenance action.

Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.

Malayalam support: Renewable and Non-renewable sources topic purpose, working principle, components variables, performance safety checks. Technical terms English-; diagram step sequence process.

Study points

- Define Renewable and Non-renewable sources using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.
- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Renewable and Non-renewable sources is applied in photovoltaic engineering practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For ****Renewable and Non-renewable sources****, prepare a four-column revision note with **purpose**, **principle/sequence**, **important parameters**, and **application or limitation**. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Renewable and Non-renewable sources without explaining the working sequence, variables, or engineering reason for each item.

Concept and explanation

Scope. The official syllabus places **Main sources of Renewable** in this topic. The required scope is: Solar, wind, tidal, biomass and other main renewable sources.

Concept and method. Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

Aspect	What to learn
Purpose	Why the device, method, material, network, or process is required in photovoltaic engineering.
Working idea	How the input is converted, controlled, transported, stored, measured, or protected.
Performance	Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.
Engineering decision	How an engineer selects a configuration, operating method, component, or maintenance action.

Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.

Malayalam support: **Malayalam support:** lang="ml">Main sources of Renewable topic പഠിക്കേണ്ട വിഷയം purpose, working principle, പഠിക്കേണ്ട components പഠിക്കേണ്ട variables, പഠിക്കേണ്ട performance പഠിക്കേണ്ട safety checks പഠിക്കേണ്ട പഠിക്കേണ്ട. Technical terms English-പഠിക്കേണ്ട പഠിക്കേണ്ട; diagram പഠിക്കേണ്ട step sequence പഠിക്കേണ്ട process പഠിക്കേണ്ട പഠിക്കേണ്ട.

Study points

- Define Main sources of Renewable using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.
- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Main sources of Renewable is applied in photovoltaic engineering practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For ****Main sources of Renewable****, prepare a four-column revision note with **purpose**, **principle/sequence**, **important parameters**, and **application or limitation**. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Main sources of Renewable without explaining the working sequence, variables, or engineering reason for each item.

Concept and explanation

Scope. The official syllabus places **Applications, Advantages & Disadvantages of Renewable Energy** in this topic. The required scope is: Applications, advantages and disadvantages of renewable energy.

Concept and method. Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

Aspect	What to learn
Purpose	Why the device, method, material, network, or process is required in photovoltaic engineering.
Working idea	How the input is converted, controlled, transported, stored, measured, or protected.
Performance	Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.
Engineering decision	How an engineer selects a configuration, operating method, component, or maintenance action.

Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.

Malayalam support: Applications, Advantages & Disadvantages of Renewable Energy topic purpose, working principle, components variables, performance safety checks diagram step sequence process.

Study points

- Define Applications, Advantages & Disadvantages of Renewable Energy using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.

- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Applications, Advantages & Disadvantages of Renewable Energy is applied in photovoltaic engineering practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For ****Applications, Advantages & Disadvantages of Renewable Energy****, prepare a four-column revision note with ***purpose***, ***principle/sequence***, ***important parameters***, and ***application or limitation***. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Applications, Advantages & Disadvantages of Renewable Energy without explaining the working sequence, variables, or engineering reason for each item.

Module summary: Consolidate Renewable Energy Fundamentals through definitions, working principles, engineering applications, limitations, and examination revision.

OFFICIAL MODULE 2

Solar-Cell Fundamentals

A solar cell is a semiconductor p-n junction device that converts incident photon energy into electrical output.

Hours: 12

CO2 · Bloom level: Understanding

Learning goals

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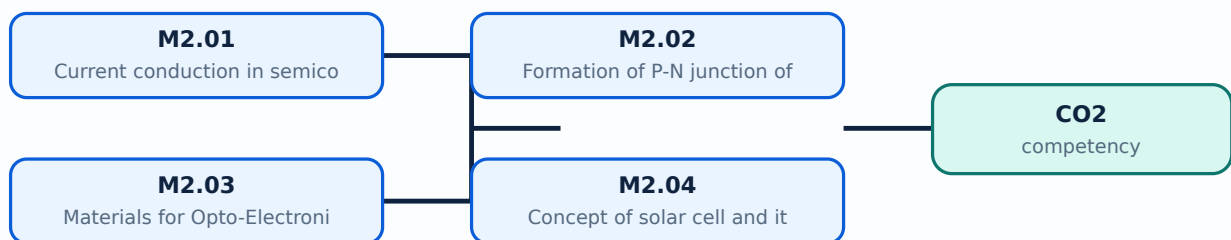
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Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-2: the listed topics build the module competency.

Concept and explanation

Scope. The official syllabus places **Current conduction in semiconductor** in this topic. The required scope is: Current conduction in semiconductor.

Concept and method. Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

Study frame for Current conduction in semiconductor	
Aspect	What to learn
Purpose	Why the device, method, material, network, or process is required in solar-cell technology.
Working idea	How the input is converted, controlled, transported, stored, measured, or protected.
Performance	Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.
Engineering decision	How an engineer selects a configuration, operating method, component, or maintenance action.

Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.

Malayalam support: **Malayalam support:** Current conduction in semiconductor
 topic
 purpose, working principle, components variables, performance
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 . Technical terms
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Study points

- Define Current conduction in semiconductor using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.
- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Current conduction in semiconductor is applied in solar-cell technology practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For **Current conduction in semiconductor**, prepare a four-column revision note with **purpose**, **principle/sequence**, **important parameters**, and **application or limitation**. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Current conduction in semiconductor without explaining the working sequence, variables, or engineering reason for each item.

Concept and explanation

Scope. The official syllabus places **Formation of P-N junction of Semiconductor** in this topic. The required scope is: Formation of p-n junction in semiconductor.

Concept and method. Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

Aspect	What to learn
Purpose	Why the device, method, material, network, or process is required in solar-cell technology.
Working idea	How the input is converted, controlled, transported, stored, measured, or protected.
Performance	Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.
Engineering decision	How an engineer selects a configuration, operating method, component, or maintenance action.

Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.

Malayalam support: Malayalam support: Formation of P-N junction of Semiconductor താഴെ വിഷയം പരിശോധിക്കുക. ഉദ്ദേശ്യം, പ്രവർത്തന തത്വം, ഘടകങ്ങൾ, വ്യക്തിഗത വേരിയബിൾ, പ്രവർത്തന പ്രകാരം സുരക്ഷാ പരിശോധനകൾ താഴെ പരിശോധിക്കുക. Technical terms English-ൽ പരിശോധിക്കുക; diagram പരിശോധിക്കുക step sequence പരിശോധിക്കുക process പരിശോധിക്കുക.

Study points

- Define Formation of P-N junction of Semiconductor using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.
- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Formation of P-N junction of Semiconductor is applied in solar-cell technology practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For **Formation of P-N junction of Semiconductor**, prepare a four-column revision note with *purpose*, *principle/sequence*, *important parameters*, and *application or limitation*. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Formation of P-N junction of Semiconductor without explaining the working sequence, variables, or engineering reason for each item.

Concept and explanation

Scope. The official syllabus places **Materials for Opto-Electronic applications** in this topic. The required scope is: Materials for optoelectronic applications.

Concept and method. Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

Aspect	What to learn
Purpose	Why the device, method, material, network, or process is required in solar-cell technology.
Working idea	How the input is converted, controlled, transported, stored, measured, or protected.
Performance	Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.
Engineering decision	How an engineer selects a configuration, operating method, component, or maintenance action.

Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.

Malayalam support: **Malayalam support:** Materials for Opto-Electronic applications ഈ വിഷയത്തിൽ ഉൾപ്പെട്ട പദങ്ങൾ purpose, working principle, ഈ വിഷയത്തിൽ ഉൾപ്പെട്ട components ഈ വിഷയത്തിൽ ഉൾപ്പെട്ട variables, ഈ വിഷയത്തിൽ ഉൾപ്പെട്ട performance ഈ വിഷയത്തിൽ ഉൾപ്പെട്ട safety checks ഈ വിഷയത്തിൽ ഉൾപ്പെട്ട Technical terms English- ഈ വിഷയത്തിൽ ഉൾപ്പെട്ട; diagram ഈ വിഷയത്തിൽ ഉൾപ്പെട്ട step sequence ഈ വിഷയത്തിൽ ഉൾപ്പെട്ട process ഈ വിഷയത്തിൽ ഉൾപ്പെട്ട.

Study points

- Define Materials for Opto-Electronic applications using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.
- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Materials for Opto-Electronic applications is applied in solar-cell technology practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For **Materials for Opto-Electronic applications**, prepare a four-column revision note with **purpose**, **principle/sequence**, **important parameters**, and **application or limitation**. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Materials for Opto-Electronic applications without explaining the working sequence, variables, or engineering reason for each item.

Concept and explanation

Scope. The official syllabus places **Concept of solar cell and its main elements** in this topic. The required scope is: Solar-cell concept and main elements.

Concept and method. Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

Aspect	What to learn
Purpose	Why the device, method, material, network, or process is required in solar-cell technology.
Working idea	How the input is converted, controlled, transported, stored, measured, or protected.
Performance	Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.
Engineering decision	How an engineer selects a configuration, operating method, component, or maintenance action.

Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.

Malayalam support: Malayalam support: Concept of solar cell and its main elements topic purpose, working principle, components variables, performance safety checks Technical terms English- diagram step sequence process

Study points

- Define Concept of solar cell and its main elements using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.
- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Concept of solar cell and its main elements is applied in solar-cell technology practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For **Concept of solar cell and its main elements**, prepare a four-column revision note with **purpose**, **principle/sequence**, **important parameters**, and **application or limitation**. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Concept of solar cell and its main elements without explaining the working sequence, variables, or engineering reason for each item.

Module summary: Consolidate Solar-Cell Fundamentals through definitions, working principles, engineering applications, limitations, and examination revision.

OFFICIAL MODULE 3

Solar-Cell Characteristics and Efficiency

Cell performance is described by its current-voltage characteristic and parameters such as short-circuit current, open-circuit voltage, fill factor, maximum-power point, and efficiency.

Hours: 16

CO3 · Bloom level: Understanding

Learning goals

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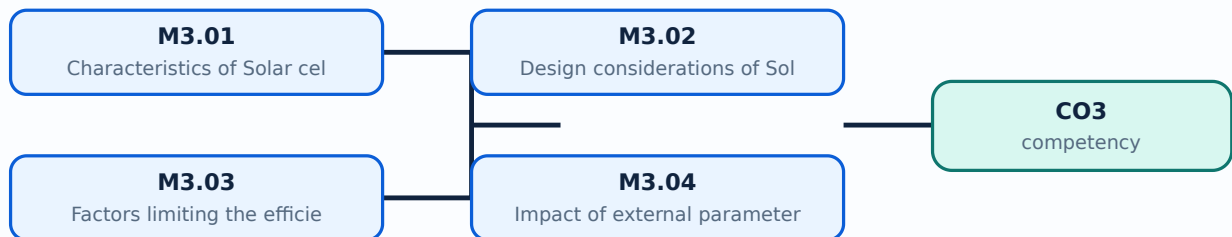
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Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-3: the listed topics build the module competency.

Concept and explanation

Scope. The official syllabus places **Characteristics of Solar cells** in this topic. The required scope is: Solar-cell characteristics.

Concept and method. Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

Aspect	What to learn
Purpose	Why the device, method, material, network, or process is required in solar-cell technology.
Working idea	How the input is converted, controlled, transported, stored, measured, or protected.
Performance	Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.
Engineering decision	How an engineer selects a configuration, operating method, component, or maintenance action.

Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.

Malayalam support: **Characteristics of Solar cells** എന്നു് വിഷയം. പഠിക്കേണ്ടതു് **purpose, working principle, components, variables, performance, safety checks** എന്നിവ. **Technical terms** English-ൽ പഠിക്കേണ്ടതു്; **diagram** എന്നതു് **step sequence** എന്നതു് **process** എന്നതു്.

Study points

- Define Characteristics of Solar cells using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.
- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Characteristics of Solar cells is applied in solar-cell technology practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For ****Characteristics of Solar cells****, prepare a four-column revision note with **purpose**, **principle/sequence**, **important parameters**, and **application or limitation**. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Characteristics of Solar cells without explaining the working sequence, variables, or engineering reason for each item.

Concept and explanation

Scope. The official syllabus places **Design considerations of Solar cells** in this topic. The required scope is: Solar-cell design considerations.

Concept and method. Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

Aspect	What to learn
Purpose	Why the device, method, material, network, or process is required in solar-cell technology.
Working idea	How the input is converted, controlled, transported, stored, measured, or protected.
Performance	Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.
Engineering decision	How an engineer selects a configuration, operating method, component, or maintenance action.

Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.

Malayalam support: Design considerations of Solar cells topic purpose, working principle, components variables, performance safety checks. Technical terms English-; diagram step sequence process.

Study points

- Define Design considerations of Solar cells using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.
- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Design considerations of Solar cells is applied in solar-cell technology practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For ****Design considerations of Solar cells****, prepare a four-column revision note with **purpose**, **principle/sequence**, **important parameters**, and **application or limitation**. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Design considerations of Solar cells without explaining the working sequence, variables, or engineering reason for each item.

Concept and explanation

Scope. The official syllabus places **Factors limiting the efficiency of solar cell** in this topic. The required scope is: Factors limiting solar-cell efficiency.

Concept and method. Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

Aspect	What to learn
Purpose	Why the device, method, material, network, or process is required in solar-cell technology.
Working idea	How the input is converted, controlled, transported, stored, measured, or protected.
Performance	Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.
Engineering decision	How an engineer selects a configuration, operating method, component, or maintenance action.

Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.

Malayalam support: **Malayalam support:** Factors limiting the efficiency of solar cell
 topic
 purpose, working principle, components variables, performance
 safety checks
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 diagram
 step sequence
 process
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Study points

- Define Factors limiting the efficiency of solar cell using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.
- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Factors limiting the efficiency of solar cell is applied in solar-cell technology practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For ****Factors limiting the efficiency of solar cell****, prepare a four-column revision note with ***purpose***, ***principle/sequence***, ***important parameters***, and ***application or limitation***. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Factors limiting the efficiency of solar cell without explaining the working sequence, variables, or engineering reason for each item.

Concept and explanation

Scope. The official syllabus places **Impact of external parameters on solar cell performances** in this topic. The required scope is: Impact of external parameters on solar-cell performance.

Concept and method. Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

Study frame for Impact of external parameters on solar cell performances	
Aspect	What to learn
Purpose	Why the device, method, material, network, or process is required in solar-cell technology.
Working idea	How the input is converted, controlled, transported, stored, measured, or protected.
Performance	Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.
Engineering decision	How an engineer selects a configuration, operating method, component, or maintenance action.

Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.

Malayalam support: **Malayalam support:** Impact of external parameters on solar cell performances topic
purpose, working principle, components
variables, performance safety checks English-
 Technical terms English-
 diagram step
 sequence process.

Study points

- Define Impact of external parameters on solar cell performances using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.
- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Impact of external parameters on solar cell performances is applied in solar-cell technology practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For ****Impact of external parameters on solar cell performances****, prepare a four-column revision note with ***purpose***, ***principle/sequence***, ***important parameters***, and ***application or limitation***. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Impact of external parameters on solar cell performances without explaining the working sequence, variables, or engineering reason for each item.

Module summary: Consolidate Solar-Cell Characteristics and Efficiency through definitions, working principles, engineering applications, limitations, and examination revision.

OFFICIAL MODULE 4

Solar-Cell and PV-Module Fabrication

Cells are classified and interconnected into modules, panels, and arrays to obtain usable voltage, current, mechanical protection, and field reliability.

Hours: 18

CO4 · Bloom level: Understanding

Learning goals

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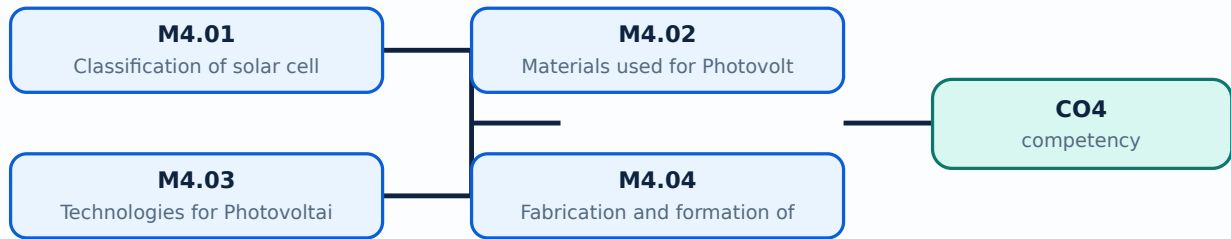
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Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-4: the listed topics build the module competency.

Engineering / workplace application: Classification of solar cells is applied in photovoltaic fabrication practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For ****Classification of solar cells****, prepare a four-column revision note with **purpose**, **principle/sequence**, **important parameters**, and **application or limitation**. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Classification of solar cells without explaining the working sequence, variables, or engineering reason for each item.

Concept and explanation

Scope. The official syllabus places **Materials used for Photovoltaic Cells** in this topic. The required scope is: Single-crystalline and polycrystalline silicon; amorphous silicon, GaAs, CdTe, CIGS, DSSC, organic and quantum-dot materials.

Concept and method. Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

Aspect	What to learn
Purpose	Why the device, method, material, network, or process is required in photovoltaic fabrication.
Working idea	How the input is converted, controlled, transported, stored, measured, or protected.
Performance	Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.
Engineering decision	How an engineer selects a configuration, operating method, component, or maintenance action.

Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.

Malayalam support: **Malayalam support:** Materials used for Photovoltaic Cells topic purpose, working principle, components variables, performance safety checks English- diagram step sequence process

Study points

- Define Materials used for Photovoltaic Cells using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.
- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Materials used for Photovoltaic Cells is applied in photovoltaic fabrication practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For ****Materials used for Photovoltaic Cells****, prepare a four-column revision note with **purpose**, **principle/sequence**, **important parameters**, and **application or limitation**. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Materials used for Photovoltaic Cells without explaining the working sequence, variables, or engineering reason for each item.

Concept and explanation

Scope. The official syllabus places **Technologies for Photovoltaic Cells Fabrication** in this topic. The required scope is: Photovoltaic-cell fabrication technologies.

Concept and method. Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

Aspect	What to learn
Purpose	Why the device, method, material, network, or process is required in photovoltaic fabrication.
Working idea	How the input is converted, controlled, transported, stored, measured, or protected.
Performance	Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.
Engineering decision	How an engineer selects a configuration, operating method, component, or maintenance action.

Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.

Malayalam support: **Technologies for Photovoltaic Cells Fabrication** topic purpose, working principle, components variables, performance safety checks. Technical terms English-; diagram step sequence process.

Study points

- Define Technologies for Photovoltaic Cells Fabrication using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.
- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Technologies for Photovoltaic Cells Fabrication is applied in photovoltaic fabrication practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For **Technologies for Photovoltaic Cells Fabrication**, prepare a four-column revision note with **purpose**, **principle/sequence**, **important parameters**, and **application or limitation**. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Technologies for Photovoltaic Cells Fabrication without explaining the working sequence, variables, or engineering reason for each item.

Concept and explanation

Scope. The official syllabus places **Fabrication and formation of PV modules** in this topic. The required scope is: PV cell, module, panel and array concepts and formation; silicon and non-silicon groups.

Concept and method. Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

Aspect	What to learn
Purpose	Why the device, method, material, network, or process is required in photovoltaic fabrication.
Working idea	How the input is converted, controlled, transported, stored, measured, or protected.
Performance	Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.
Engineering decision	How an engineer selects a configuration, operating method, component, or maintenance action.

Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.

Malayalam support: **Malayalam support:** lang="ml">Fabrication and formation of PV modules topic പരീക്ഷാപരമായ പരിചയം purpose, working principle, പരീക്ഷാ components പരിചയം variables, പരിചയം performance പരിചയം safety checks പരിചയം പരിചയം പരിചയം. Technical terms English-പരിചയം പരിചയം; diagram പരിചയം പരിചയം step sequence പരിചയം പരിചയം പരിചയം പരിചയം പരിചയം പരിചയം.

Study points

- Define Fabrication and formation of PV modules using the official syllabus wording.
- Explain the sequence or cause-and-effect relationship before listing details.
- Use a labelled diagram, comparison, unit, or operating condition wherever it improves the answer.
- Connect the topic to a genuine engineering application and one possible fault, limitation, or safety concern.

Engineering / workplace application: Fabrication and formation of PV modules is applied in photovoltaic fabrication practice, design review, commissioning, troubleshooting, or examination analysis.

Illustrative example: Study example: For **Fabrication and formation of PV modules**, prepare a four-column revision note with **purpose**, **principle/sequence**, **important parameters**, and **application or limitation**. Use the official scope above as the checklist and verify that no listed term is omitted.

Common mistake: A common mistake is to write only a list for Fabrication and formation of PV modules without explaining the working sequence, variables, or engineering reason for each item.

Module summary: Consolidate Solar-Cell and PV-Module Fabrication through definitions, working principles, engineering applications, limitations, and examination revision.

Formula and Notation Bank

Formula note: The supplied syllabus is primarily procedural or conceptual and does not prescribe a compulsory numerical formula bank. Focus on the official terminology, sequences, records, and practical demonstrations listed in the modules.

Diagram Library for Rapid Revision

	
	Module 2: Solar-Cell
	Module 3: Solar-Cell
	Module 4: Solar-Cell and PV-
	

Official Activities and Practical Requirements

Official activities / self-learning

- The supplied syllabus does not provide a separate official self-learning activity list for this theory course. Use the topic procedures, worked examples, diagrams, and module questions in this lesson for structured study.

Practical requirements / equipment

- The supplied syllabus does not prescribe separate practical equipment for this theory course.

Assessment Methodology

The supplied PDF does not specify a separate CIE/ESE distribution.

- The supplied PDF lists Series Test – I and Series Test – II entries, but a complete official CIE/ESE mark distribution and question-paper pattern are not specified in the supplied PDF.
- The practice model paper in this lesson is therefore explicitly a study aid, not an official examination paper.

Module-wise Questions and Answers

module-1 — Renewable Energy Fundamentals

1. Explain Need of switching to Renewable Energy sources. [3 marks]

Scope. The official syllabus places *Need of switching to Renewable Energy sources* in this topic. The required scope is: Need for switching to renewable energy sources.

Concept and method. Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

<caption>Study frame for Need of switching to Renewable Energy sources</caption>

<thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in photovoltaic engineering.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

2. Explain Renewable and Non-renewable sources. [3 marks]

<p>Scope. The official syllabus places Renewable and Non-renewable sources in this topic. The required scope is: Difference between renewable and non-renewable sources.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Renewable and Non-renewable sources</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in photovoltaic engineering.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

3. Explain Main sources of Renewable. [3 marks]

Scope. The official syllabus places **Main sources of Renewable** in this topic. The required scope is: Solar, wind, tidal, biomass and other main renewable sources.

Concept and method. Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

Aspect	What to learn
Purpose	Why the device, method, material, network, or process is required in photovoltaic engineering.
Working idea	How the input is converted, controlled, transported, stored, measured, or protected.
Performance	Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.
Engineering decision	How an engineer selects a configuration, operating method, component, or maintenance action.

Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

4. Explain Applications, Advantages & Disadvantages of Renewable Energy. [3 marks]

Scope. The official syllabus places **Applications, Advantages & Disadvantages of Renewable Energy** in this topic. The required scope is: Applications, advantages and disadvantages of renewable energy.

Concept and method. Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an

examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

</p> <div class="table-wrap"><table><caption>Study frame for Applications, Advantages & Disadvantages of Renewable Energy</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in photovoltaic engineering.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-2 — Solar-Cell Fundamentals

5. Explain Current conduction in semiconductor. [3 marks]

<p>Scope. The official syllabus places Current conduction in semiconductor in this topic. The required scope is: Current conduction in semiconductor.</p> <p>Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.</p> <div class="table-wrap"><table><caption>Study frame for Current conduction in semiconductor</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in solar-cell technology.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same

module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

6. Explain Formation of P-N junction of Semiconductor. [3 marks]

Scope. The official syllabus places **Formation of P-N junction of Semiconductor** in this topic. The required scope is: Formation of p-n junction in semiconductor.

Concept and method. Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

Aspect	What to learn
Purpose	Why the device, method, material, network, or process is required in solar-cell technology.
Working idea	How the input is converted, controlled, transported, stored, measured, or protected.
Performance	Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.
Engineering decision	How an engineer selects a configuration, operating method, component, or maintenance action.

Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

7. Explain Materials for Opto-Electronic applications. [3 marks]

Scope. The official syllabus places **Materials for Opto-Electronic applications** in this topic. The required scope is: Materials for optoelectronic applications.

Concept and method. Study this topic as a

sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

Aspect	What to learn
Purpose	Why the device, method, material, network, or process is required in solar-cell technology.
Working idea	How the input is converted, controlled, transported, stored, measured, or protected.
Performance	Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.
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Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

8. Explain Concept of solar cell and its main elements. [3 marks]

Scope. The official syllabus places *Concept of solar cell and its main elements* in this topic. The required scope is: Solar-cell concept and main elements.

Concept and method. Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

Aspect	What to learn
Purpose	Why the device, method, material, network, or process is required in solar-cell technology.
Working idea	How the input is converted, controlled, transported, stored, measured, or protected.
Performance	Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.
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Technical

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Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-3 — Solar-Cell Characteristics and Efficiency

9. Explain Characteristics of Solar cells. [3 marks]

<p>Scope. The official syllabus places Characteristics of Solar cells in this topic. The required scope is: Solar-cell characteristics.</p> <p>

Concept and method. Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

</p> <div class="table-wrap"><table><caption>Study frame for Characteristics of Solar cells</caption><thead><tr><th>Aspect</th><th>What to learn</th></tr></thead><tbody><tr><th>Purpose</th><td>Why the device, method, material, network, or process is required in solar-cell technology.</td></tr><tr><th>Working idea</th><td>How the input is converted, controlled, transported, stored, measured, or protected.</td></tr><tr><th>Performance</th><td>Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.</td></tr><tr><th>Engineering decision</th><td>How an engineer selects a configuration, operating method, component, or maintenance action.</td></tr></tbody></table></div> <p>Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.</p>

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

10. Explain Design considerations of Solar cells. [3 marks]

Scope. The official syllabus places *Design considerations of Solar cells* in this topic. The required scope is: Solar-cell design considerations.

Concept and method. Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

Aspect	What to learn
Purpose	Why the device, method, material, network, or process is required in solar-cell technology.
Working idea	How the input is converted, controlled, transported, stored, measured, or protected.
Performance	Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.
Engineering decision	How an engineer selects a configuration, operating method, component, or maintenance action.

Technical treatment. Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

11. Explain Factors limiting the efficiency of solar cell. [3 marks]

Scope. The official syllabus places *Factors limiting the efficiency of solar cell* in this topic. The required scope is: Factors limiting solar-cell efficiency.

Concept and method. Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

Aspect	What to learn
Purpose	Why the device, method, material, network, or process is required in solar-cell technology.
Working idea	How the input is converted, controlled, transported, stored, measured, or protected.

Performance

Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.

Engineering decision

How an engineer selects a configuration, operating method, component, or maintenance action.

Technical treatment.

Relate the named subtopic to the neighbouring topics in the same module. A good answer distinguishes normal operation from fault or unsafe operation, identifies cause and effect, and ends with the practical implication. Where a calculation or comparison is expected, write the governing relationship first, define every symbol, substitute values with units, and state a clearly bounded conclusion.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

12. Explain Impact of external parameters on solar cell performances. [3 marks]

Scope.

The official syllabus places Impact of external parameters on solar cell performances in this topic. The required scope is: Impact of external parameters on solar-cell performance.

Concept and method.

Study this topic as a sequence of purpose → principle → components or variables → operation → performance or safety check. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

Aspect	What to learn
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Working idea	How the input is converted, controlled, transported, stored, measured, or protected.
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Engineering decision	How an engineer selects a configuration, operating method, component, or maintenance action.

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Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

13. Explain Classification of solar cells. [3 marks]

Scope. The official syllabus places *Classification of solar cells* in this topic. The required scope is: Classification and cell size; silicon and thin-film families.

Concept and method. Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

Aspect	What to learn
Purpose	Why the device, method, material, network, or process is required in photovoltaic fabrication.
Working idea	How the input is converted, controlled, transported, stored, measured, or protected.
Performance	Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.
Engineering decision	How an engineer selects a configuration, operating method, component, or maintenance action.

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Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

14. Explain Materials used for Photovoltaic Cells. [3 marks]

Scope. The official syllabus places *Materials used for Photovoltaic Cells* in this topic. The required scope is: Single-crystalline and polycrystalline silicon; amorphous silicon, GaAs, CdTe, CIGS, DSSC, organic and quantum-dot materials.

Concept and method. Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

Aspect	What to learn
Purpose	Why the

device, method, material, network, or process is required in photovoltaic fabrication.

Working idea	How the input is converted, controlled, transported, stored, measured, or protected.
Performance	Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.
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Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

15. Explain Technologies for Photovoltaic Cells Fabrication. [3 marks]

Scope. The official syllabus places **Technologies for Photovoltaic Cells Fabrication** in this topic. The required scope is: Photovoltaic-cell fabrication technologies.

Concept and method. Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

Aspect	What to learn
Purpose	Why the device, method, material, network, or process is required in photovoltaic fabrication.
Working idea	How the input is converted, controlled, transported, stored, measured, or protected.
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Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

16. Explain Fabrication and formation of PV modules. [3 marks]

Scope. The official syllabus places *Fabrication and formation of PV modules* in this topic. The required scope is: PV cell, module, panel and array concepts and formation; silicon and non-silicon groups.

Concept and method. Study this topic as a sequence of **purpose → principle → components or variables → operation → performance or safety check**. First define the engineering quantity or system named in the title. Then identify the input, the transformation that occurs, the output, and the condition that limits performance. In an examination answer, use the exact technical terms from the syllabus, draw a labelled block or schematic where appropriate, and state the relevant unit, assumption, or operating condition.

Study frame for Fabrication and formation of PV modules

Aspect	What to learn
Purpose	Why the device, method, material, network, or process is required in photovoltaic fabrication.
Working idea	How the input is converted, controlled, transported, stored, measured, or protected.
Performance	Which characteristic, efficiency, capacity, accuracy, reliability, safety, or environmental factor must be checked.
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Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

Practice Model Question Paper

Practice Model Paper — Not an Official Examination Paper

This paper is generated from the supplied module coverage for revision and does not claim to reproduce the official examination pattern.

1. **1.** Define or explain *Need of switching to Renewable Energy sources* with one relevant application. (5 marks)
2. **2.** Define or explain *Renewable and Non-renewable sources* with one relevant application. (5 marks)
3. **3.** Define or explain *Current conduction in semiconductor* with one relevant application. (5 marks)
4. **4.** Define or explain *Formation of P-N junction of Semiconductor* with one relevant application. (5 marks)
5. **5.** Define or explain *Characteristics of Solar cells* with one relevant application. (5 marks)
6. **6.** Define or explain *Design considerations of Solar cells* with one relevant application. (5 marks)
7. **7.** Define or explain *Classification of solar cells* with one relevant application. (5 marks)
8. **8.** Define or explain *Materials used for Photovoltaic Cells* with one relevant application. (5 marks)

Writing advice: Use headings, standard terminology, and labelled diagrams or procedures where relevant.

Quick Revision

Module-wise key points

- **Renewable Energy Fundamentals:** Consolidate Renewable Energy Fundamentals through definitions, working principles, engineering applications, limitations, and examination revision.
- **Solar-Cell Fundamentals:** Consolidate Solar-Cell Fundamentals through definitions, working principles, engineering applications, limitations, and examination revision.
- **Solar-Cell Characteristics and Efficiency:** Consolidate Solar-Cell Characteristics and Efficiency through definitions, working principles, engineering applications, limitations, and examination revision.
- **Solar-Cell and PV-Module Fabrication:** Consolidate Solar-Cell and PV-Module Fabrication through definitions, working principles, engineering applications, limitations, and examination revision.

Exam checklist

- Write the official terminology before adding explanation.

- Use a labelled diagram or step sequence when it improves the answer.
- Check conditions, boundaries, safety, hygiene, and documentation points.
- Separate official syllabus information from your own example.

Last-minute revision: Review the coverage table, formula bank, diagram library, and common-mistake notes inside every topic.

Glossary

Technical glossary

Term	Short meaning
Need of switching to Renewable Energy sources	<p>Scope. The official syllabus places Need of switching to Renewable Energy sources in this topic. The required scope is: Need for switching to renewable energy sources.</p> <p>Concept and method. Study this topic as
Renewable and Non-renewable sources	<p>Scope. The official syllabus places Renewable and Non-renewable sources in this topic. The required scope is: Difference between renewable and non-renewable sources.</p> <p>Concept and method. Study this topic as a
Main sources of Renewable	<p>Scope. The official syllabus places Main sources of Renewable in this topic. The required scope is: Solar, wind, tidal, biomass and other main renewable sources.</p> <p>Concept and method. Study this topic as a seq
Applications, Advantages & Disadvantages of Renewable Energy	<p>Scope. The official syllabus places Applications, Advantages & Disadvantages of Renewable Energy in this topic. The required scope is: Applications, advantages and disadvantages of renewable energy.</p> <p>Concept and m
Current conduction in semiconductor	<p>Scope. The official syllabus places Current conduction in semiconductor in this topic. The required scope is: Current conduction in semiconductor.</p> <p>Concept and method. Study this topic as a sequence of <stron
Formation of P-N junction of Semiconductor	<p>Scope. The official syllabus places Formation of P-N junction of Semiconductor in this topic. The required scope is: Formation of p-n junction in semiconductor.</p> <p>Concept and method. Study this topic as a sequ
Materials for Opto-Electronic applications	<p>Scope. The official syllabus places Materials for Opto-Electronic applications in this topic. The required scope is: Materials for optoelectronic applications.</p> <p>Concept and method. Study this topic as a sequ

Term	Short meaning
Concept of solar cell and its main elements	<p>Scope. The official syllabus places Concept of solar cell and its main elements in this topic. The required scope is: Solar-cell concept and main elements.</p> <p>Concept and method. Study this topic as a sequence
Characteristics of Solar cells	<p>Scope. The official syllabus places Characteristics of Solar cells in this topic. The required scope is: Solar-cell characteristics.</p> <p>Concept and method. Study this topic as a sequence of purpose → pr
Design considerations of Solar cells	<p>Scope. The official syllabus places Design considerations of Solar cells in this topic. The required scope is: Solar-cell design considerations.</p> <p>Concept and method. Study this topic as a sequence of
Factors limiting the efficiency of solar cell	<p>Scope. The official syllabus places Factors limiting the efficiency of solar cell in this topic. The required scope is: Factors limiting solar-cell efficiency.</p> <p>Concept and method. Study this topic as a seque
Impact of external parameters on solar cell performances	<p>Scope. The official syllabus places Impact of external parameters on solar cell performances in this topic. The required scope is: Impact of external parameters on solar-cell performance.</p> <p>Concept and method.
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Official References and Resources

References listed in the supplied syllabus

1. N.S. Rathore, Khobragade Chetan and Asnani Bhawana, Fundamentals of Renewable Energy Sources.
2. Mehmet Kanoğlu, Yunus A. Çengel and John M. Cimbala, Fundamentals and Applications of Renewable Energy.
3. G.N. Tiwari, Fundamentals of Renewable Energy Sources.
4. Aldo Vieira da Rosa, Fundamentals of Renewable Energy Processes.

Official learning resources listed

- <https://archive.nptel.ac.in/courses/115/107/115107116/>

In-page Search

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